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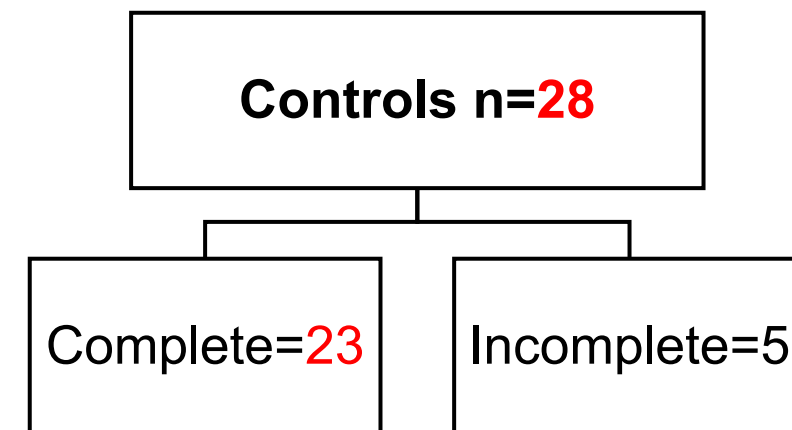
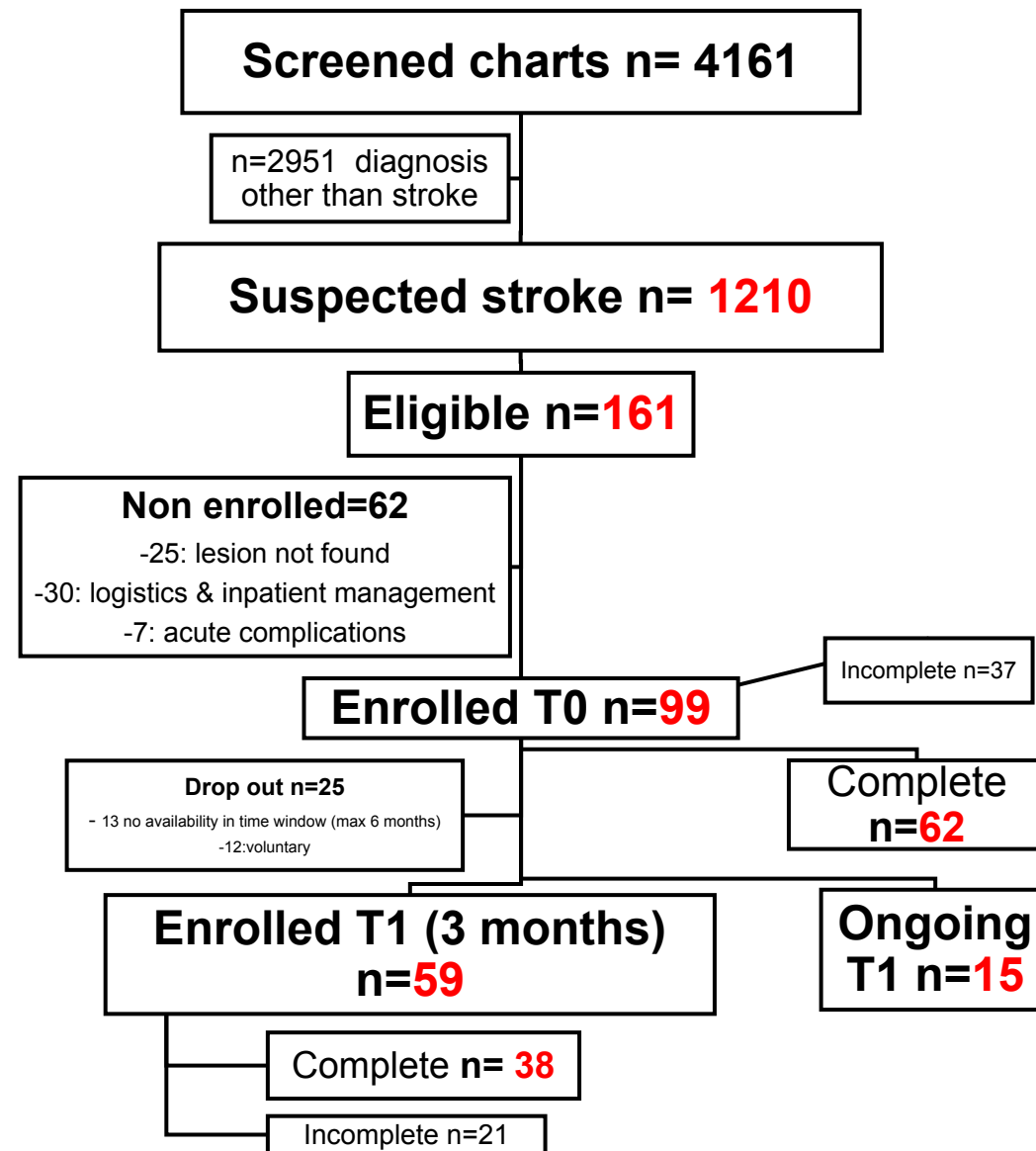


Human data acquisition update

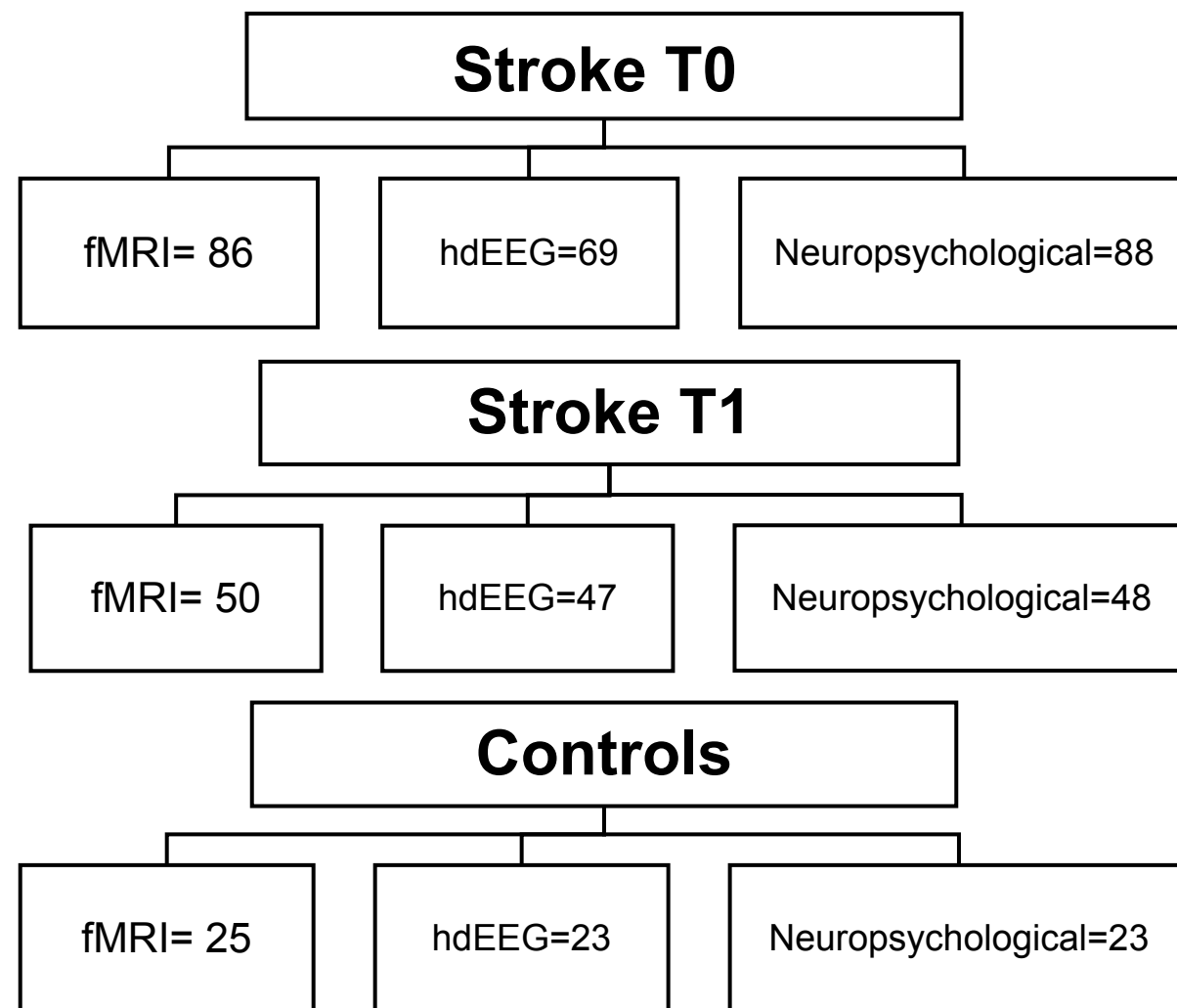
Padova Group

Nemesis Meeting Barcellona, March 26-27, 2025

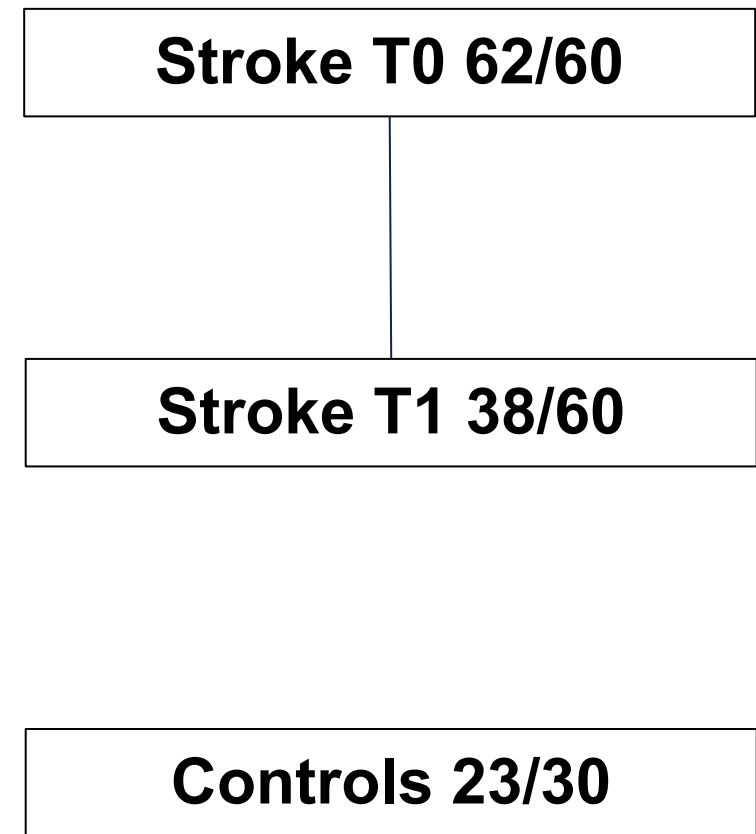
ENROLLMENT FLOWCHART



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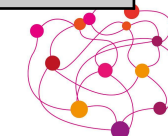


Tot.	161	139	159
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Stroke population (n=99)		
Age	Mean SD	65.1+-10.7
Gender	Male	60%
	Female	40%
Education	>13	18% (18)
	9-13	30% (30)
	<8	51% (51)
Handedness	Right	97.9% (97)
	Left	2.1% (2)
	Ambi	0% (0)
Risk factors	Smoker	37% (37)
	HBP	36% (36)
	AF	10% (10)
	CVD	8% (8)
	DM	7% (7)
	Dyslipidemia	13% (13)

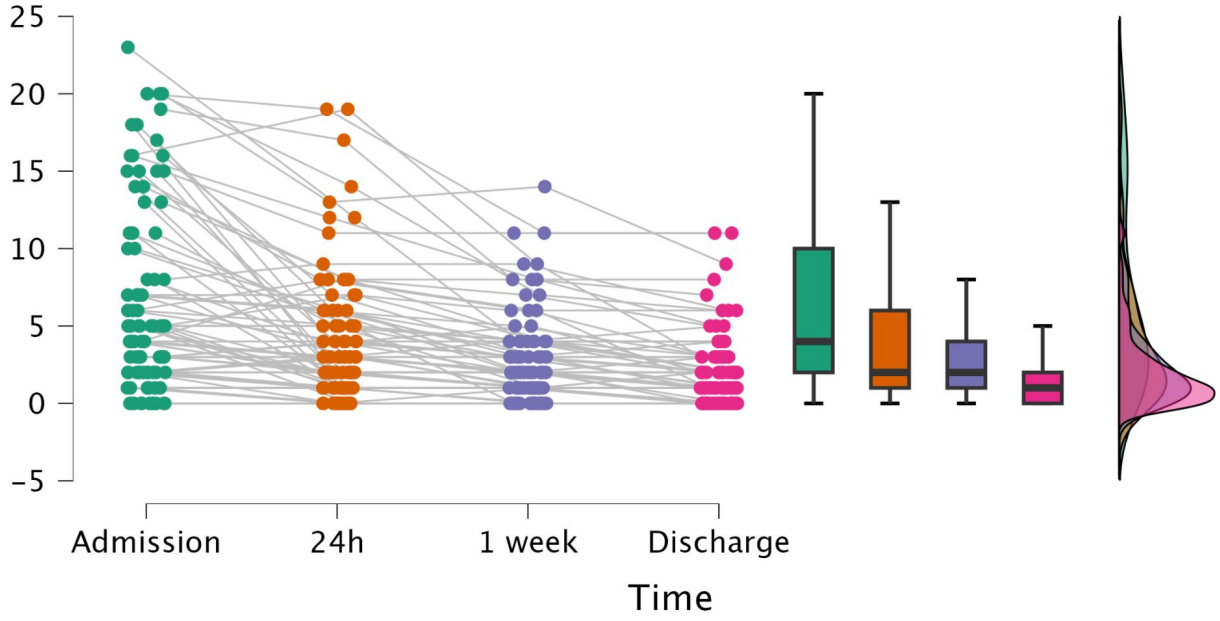
Healthy subjects (n=28)		
Age	Mean SD	63.4+-10.2
Gender	Male	50%(14)
	Female	50% (14)
Education	>13	50% (14)
	9-13	25% (7)
	<8	25 % (7)
Handedness	Right	85% (24)
	Left	3.5% (1)
	Ambi	10.7% (3)
Risk factors	Smoker	11% (3)
	HBP	21% (6)
	AF	4% (1)
	CVD	7% (2)
	DM	11% (3)
	Dyslipidemia	11% (3)



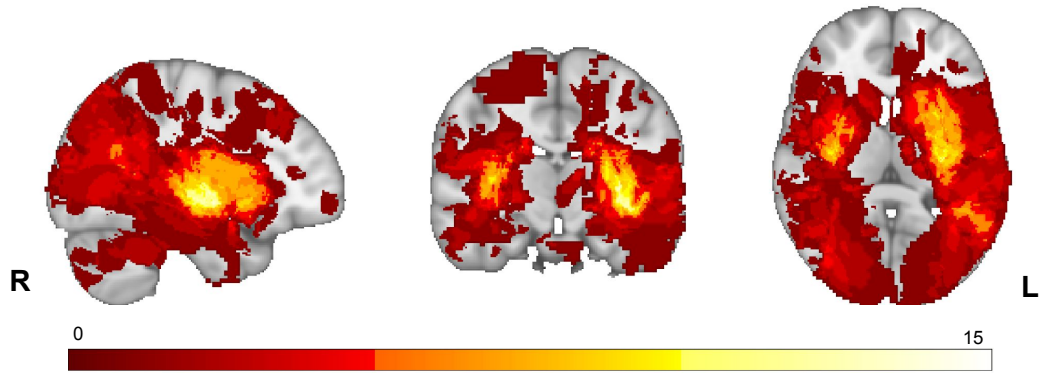
CLINICAL SAMPLE

Clinical characteristics T0 (n=99)		
preMRS	Mean SD	0.11 +- 0.1
mRS discharge	Mean	1.34 +-1
NIHSS adm	Mean	6.89 +- 6.26
NIHSS discharge	Mean	1.93 +- 2.34
Acute Complications	Major infection	7% (7)
	Minor infection	10% (10)
	Seizures	2% (2)
	Acute heart failure	2% (2)
Etiology (TOAST)	CE	29% (29)
	LAA	20% (20)
	SAO	3% (3)
	SOC	7% (7)
	SUC	31% (31)
Revascularization treatment	rTPA	17% (17)
	EVT	7% (7)
	Both	17% (17)
	None	59% (58)

NIHSS distribution during hospitalization



Lesion frequency map (n=79)



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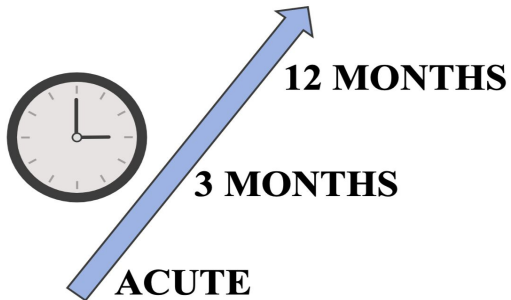


Longitudinal out-of-cohort replication of behavioural low-dimensionality in stroke

How replicable are the low-dimensional patterns of stroke behaviour ?

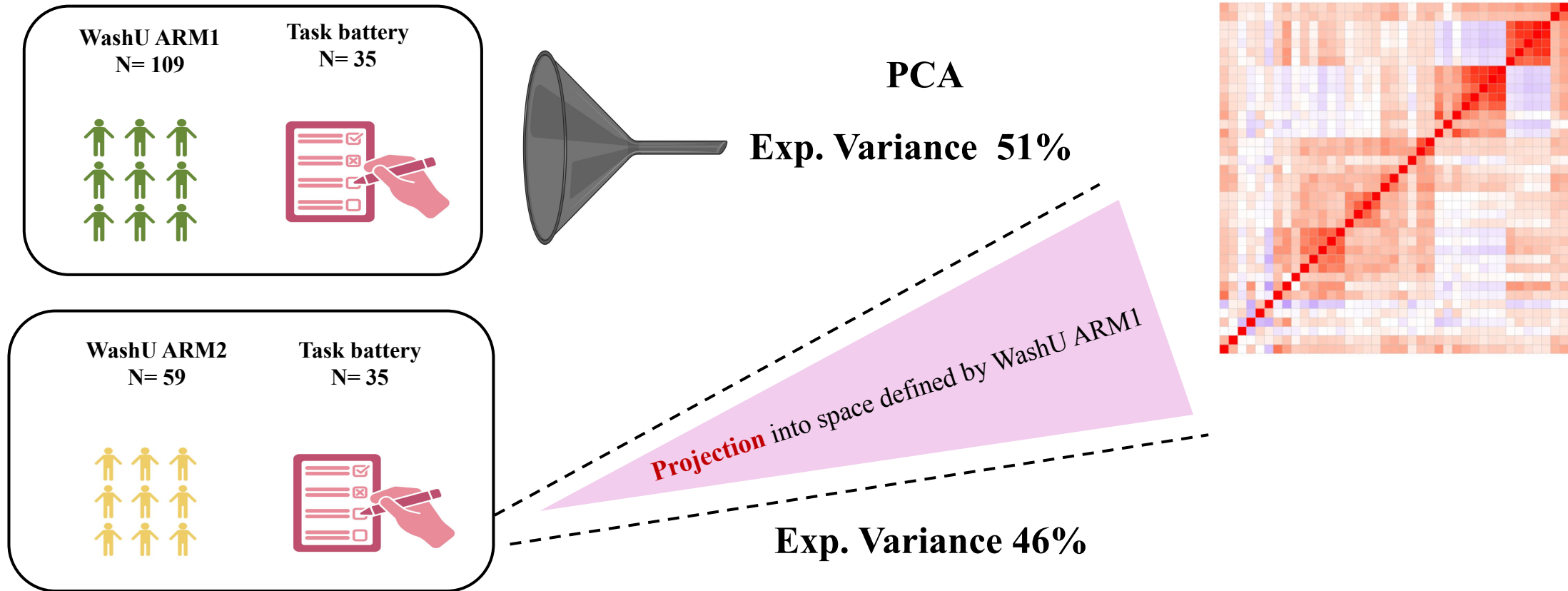
How do patients evolve along these low-dimensional patterns across recovery stages ?

How stable are their anatomical correlates across stroke recovery stages ?



How replicable is low-dimensional stroke behavioural structure?

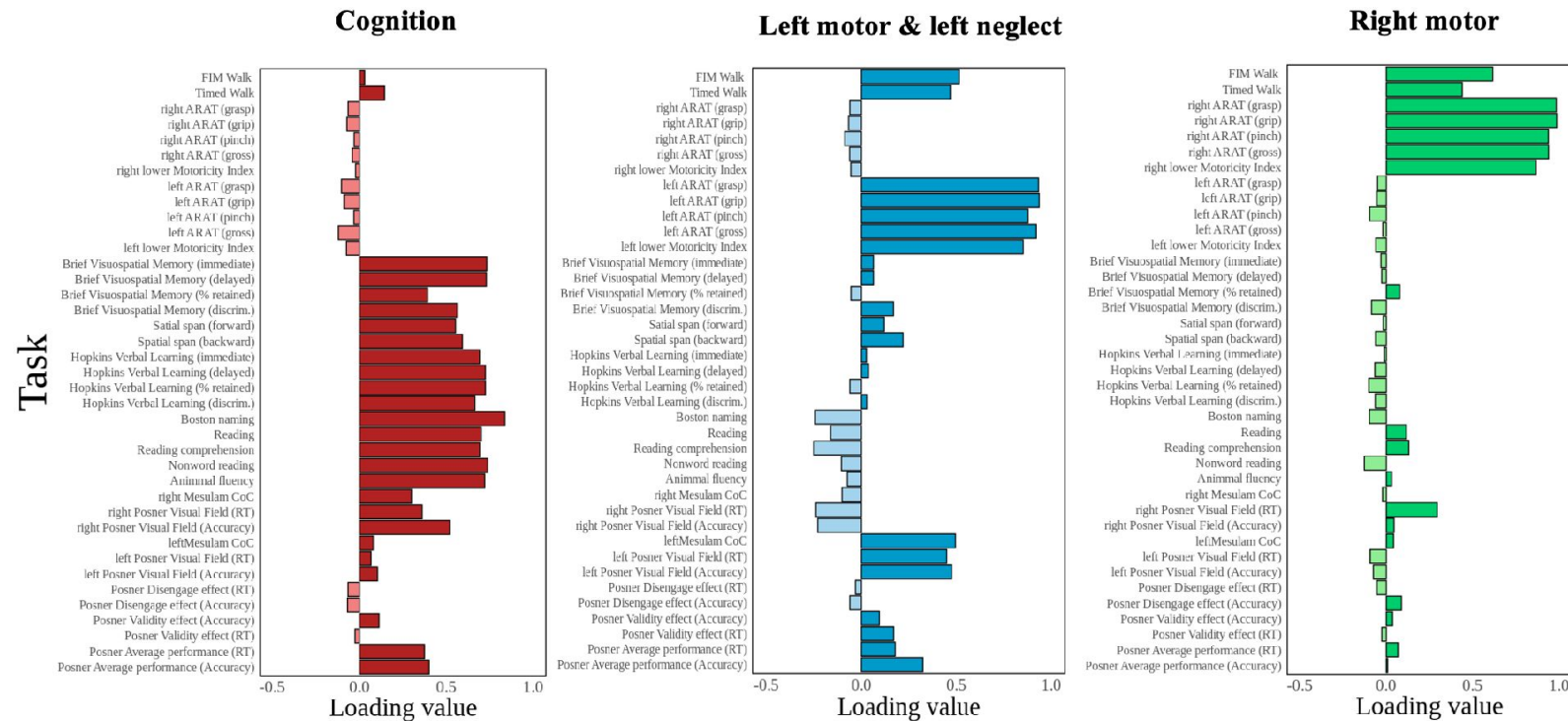
Three component solution shows strong out-of-cohort replicability



How replicable is low-dimensional stroke behavioural structure?

Stable representation of 2 motor and 1 cognitive components across cohorts

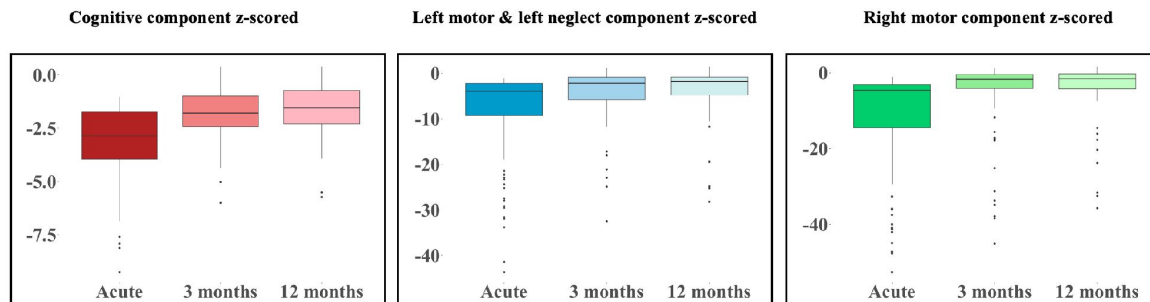
Principal components loadings



How do patients evolve along these low-dimensional patterns over time?

- Stable representation of 2 motor & 1 cognitive components across time
- Patients recover (2 weeks - 3 months) across all three components

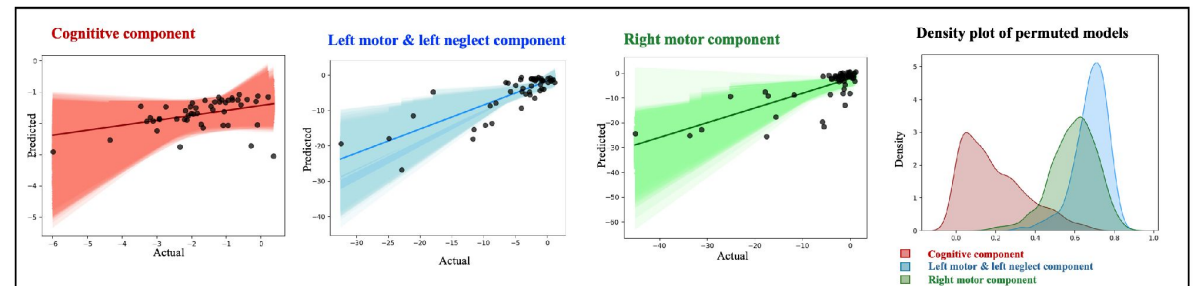
Boxplots Impaired patients (-1 sd)



- Cognitive-motor dichotomy

acute scores prediction of chronic Motor scores (59-68%) vs Cognitive scores (16%)

Bootstrapping of longitudinal regression model for each component



How stable are their anatomical correlates across stroke recovery stages ?

The cognitive-motor dichotomy mirrored in anatomy

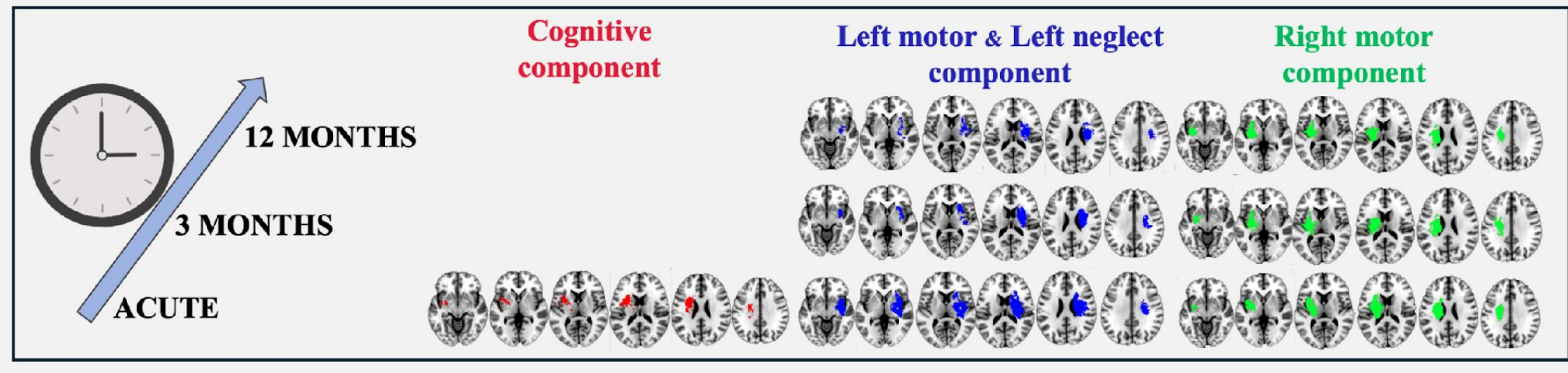
Cognitive component

structure-component decoupling over time,
beyond lesion factors shape chronic cognitive
recovery

Motor components

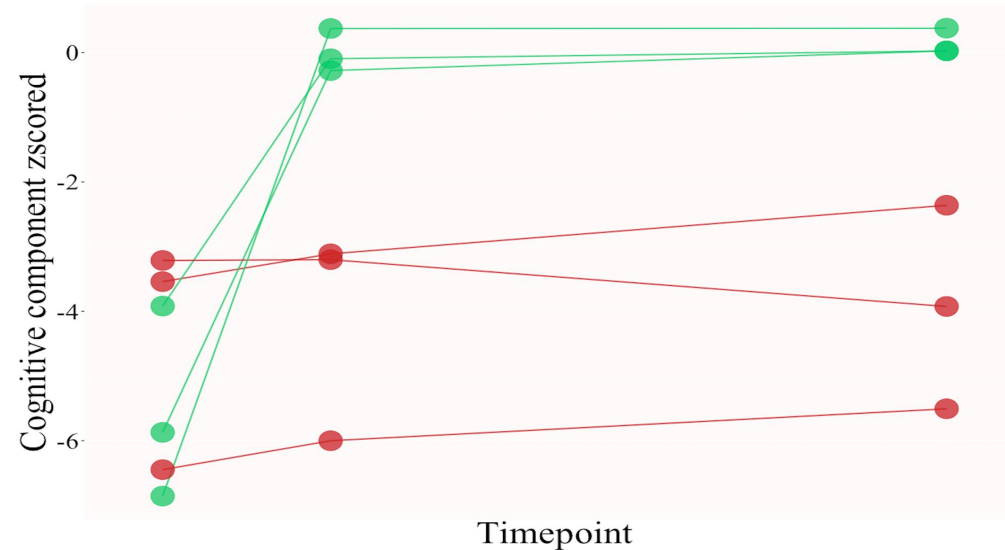
stable & time-independent anatomical relationships,
key structural regions in motor impairment and recovery

Longitudinal brain-component relationship



Key messages

- ❖ Behavioural biomarkers: replicable behavioral components across multiple cohorts (L,R motor and one cognitive component) component
- ❖ Motor recovery is strongly predicted by acute impairment and lesion location.
- ❖ In contrast, cognitive recovery is not related to acute performance and anatomy hence must be reflecting brain-wide network effects





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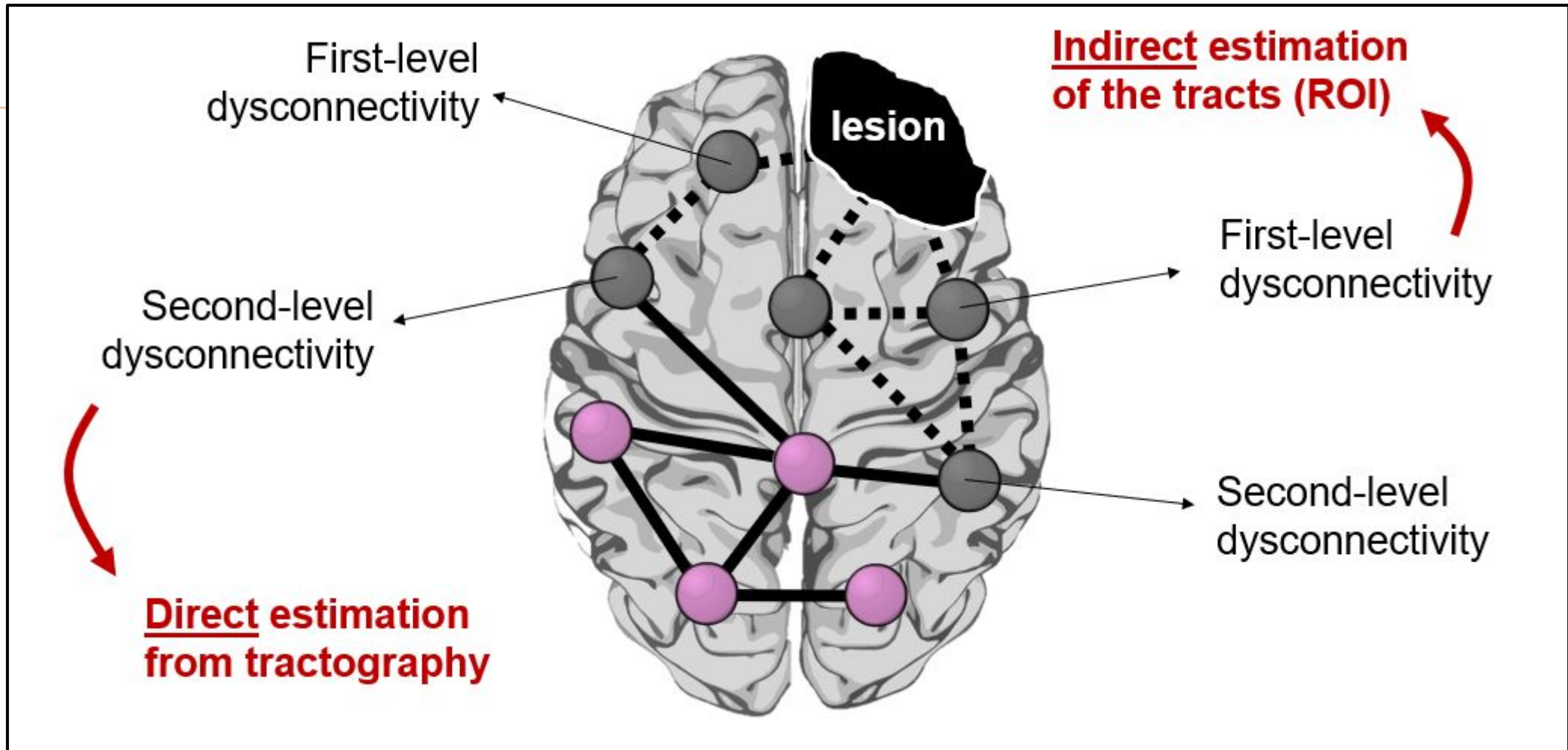


Longitudinal degeneration of microstructural and structural connectivity patterns following stroke

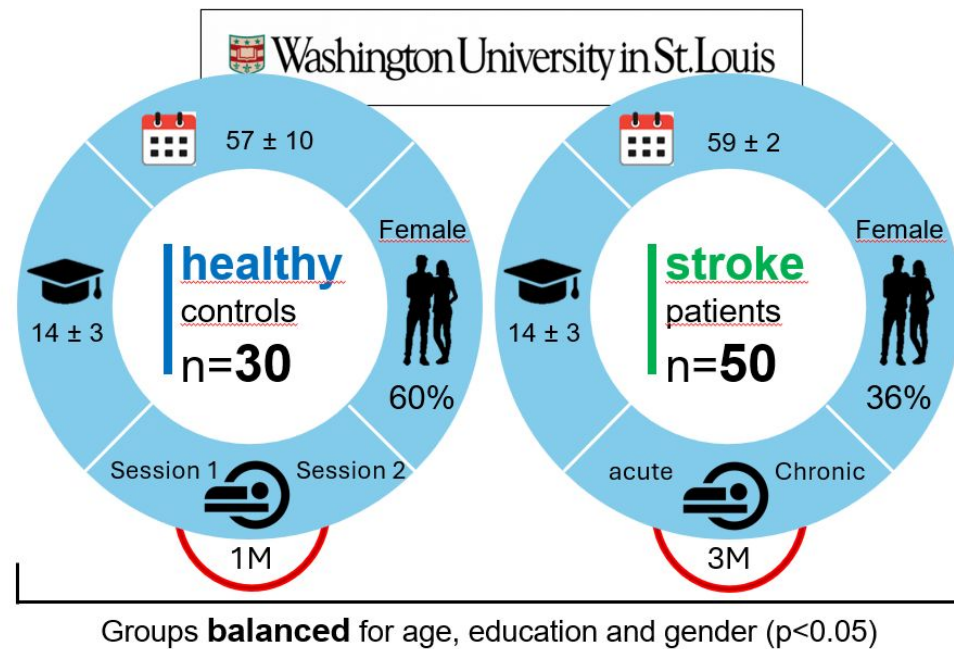
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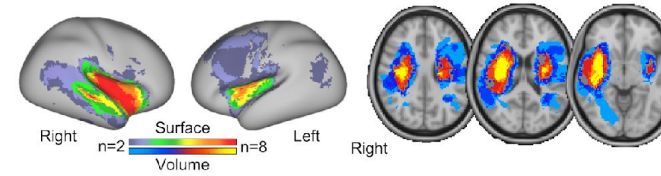
Aim: structural changes following stroke



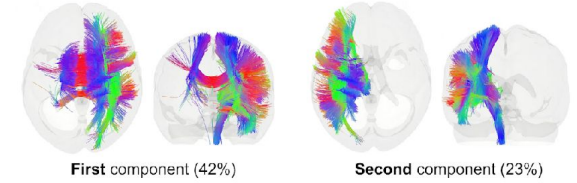
Microstructural and connectivity patterns



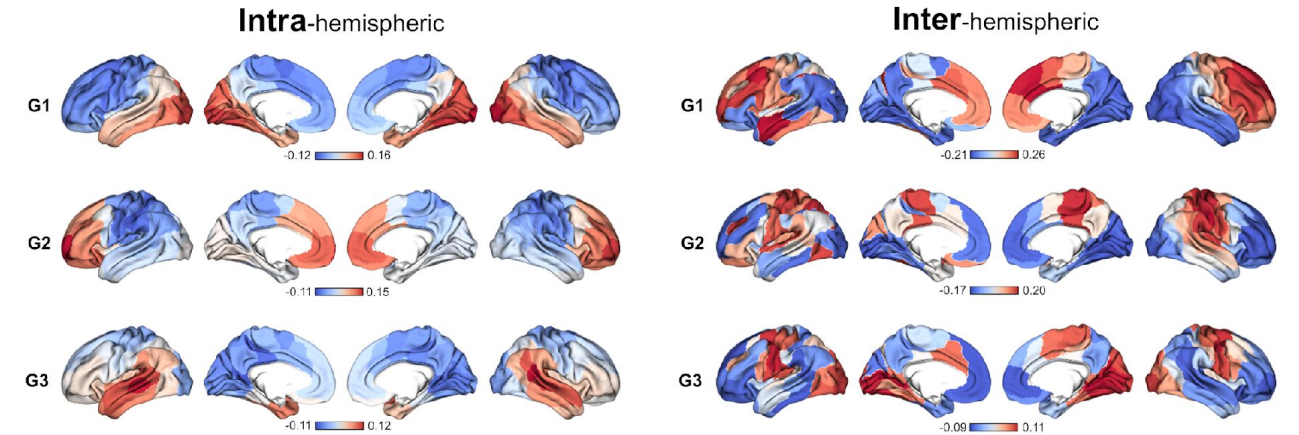
Lesion frequency map **a**



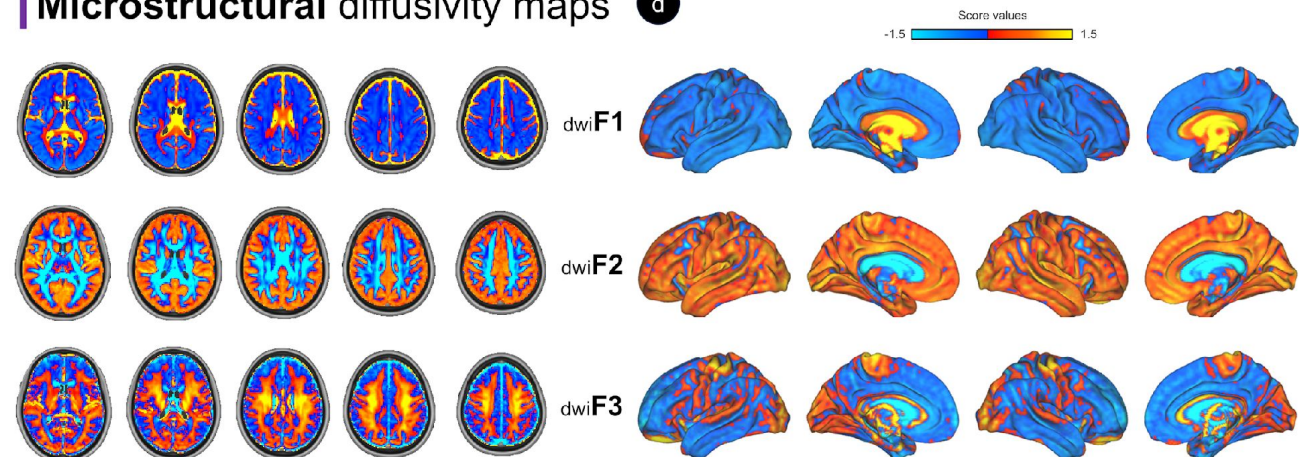
Structural disconnections **b**



Structural gradients maps **c**

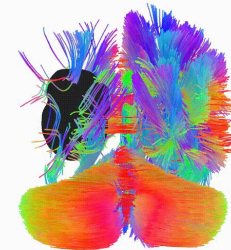


Microstructural diffusivity maps **d**



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Widespread alterations in stroke (baseline) vs controls



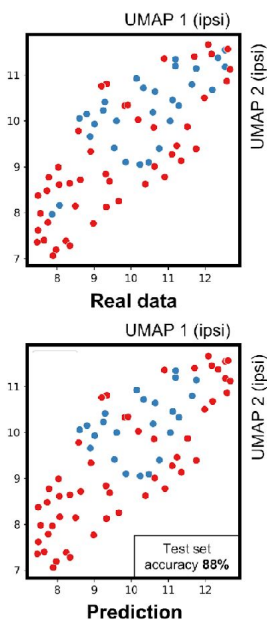
■ Healthy
 ■ Stroke (acute)
 Highly identifiable space between stroke and HC by SVM (88 - 97%)

Intra-connectivity

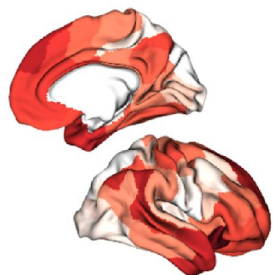
Inter-connectivity

Classification

Parcel-wise

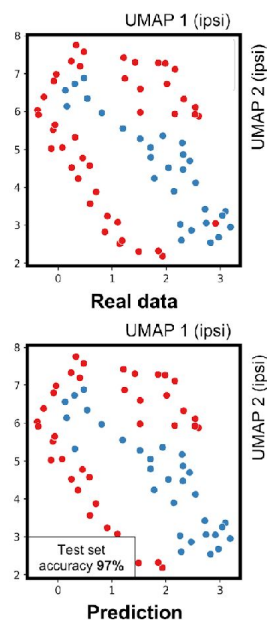


One sample
T-test
5 10

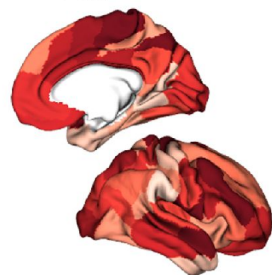


Classification

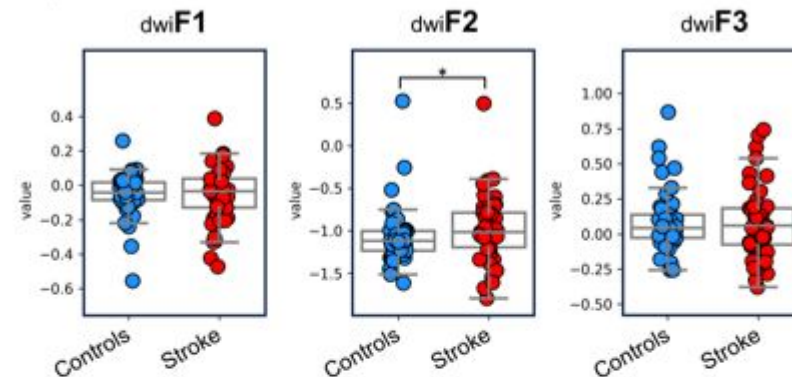
Parcel-wise



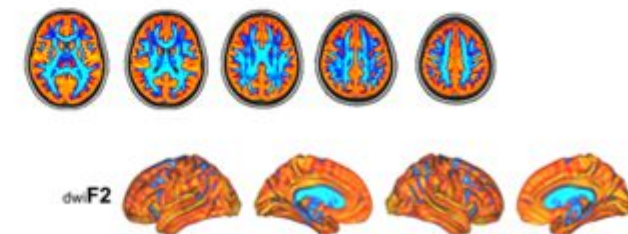
One sample
T-test
5 10



a SDC threshold 60%



Disconnected tract



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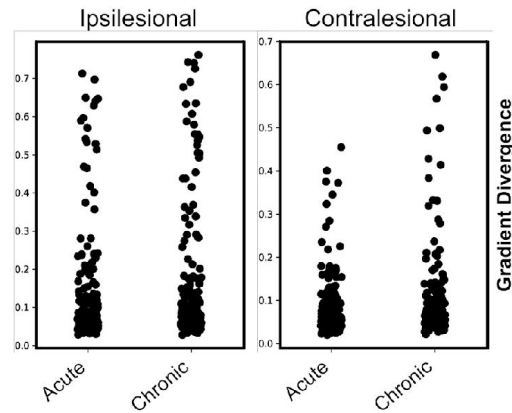


Longitudinal degeneration in stroke patients

Longitudinal differences

c

Intra-hemispheric



Inter-hemispheric

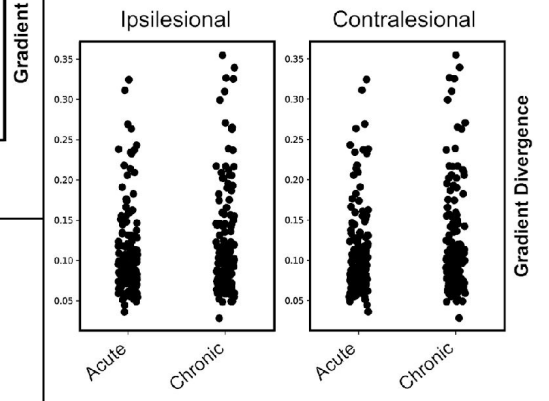
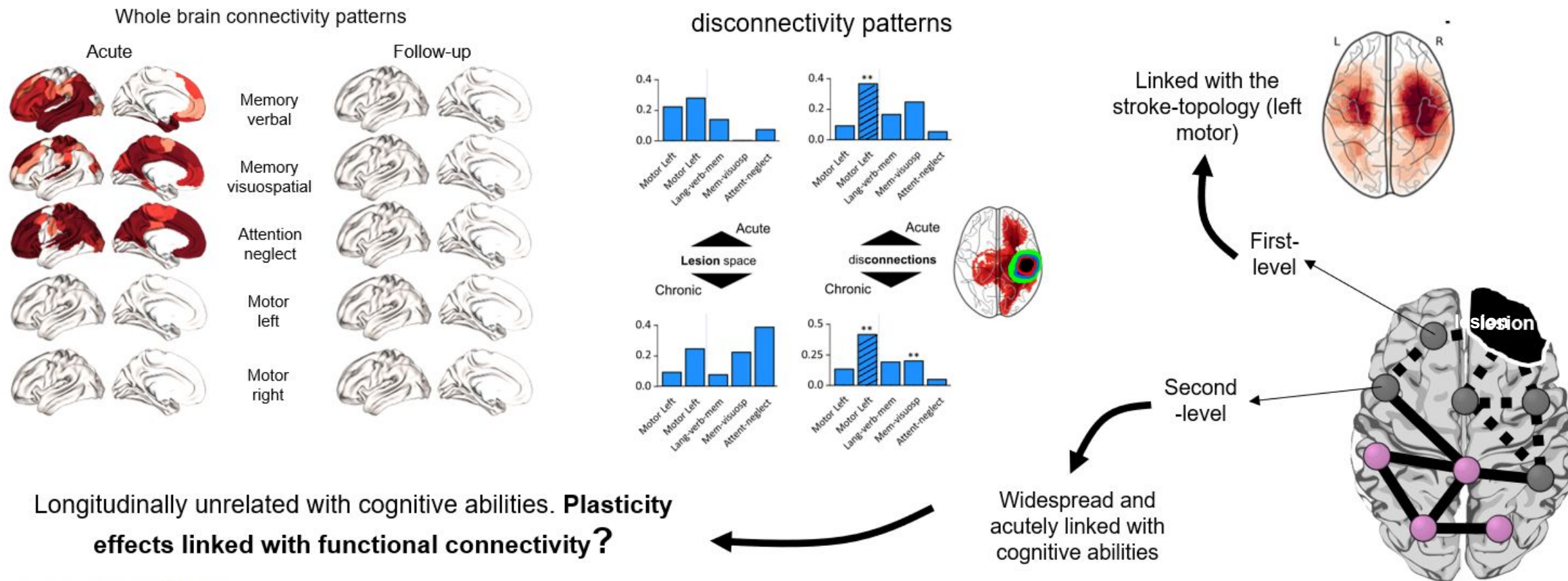


Table 2. Linear Mixed Model.

Results for ITT-L sample considering GD values at acute and 3 months follow-up.

	Estimate	95% CI	SE	T-stat	p-value
ITT-L					
Ipsilateral connectivity					
intraGD	0.024	0.010 – 0.037	0.007	3.406	0.001
interGD	0.012	0.004 – 0.020	0.004	3.035	0.003
Contralateral connectivity					
intraGD	0.018	0.008 – 0.028	0.005	3.431	0.001
interGD	0.014	0.004 – 0.025	0.005	2.667	0.008
Microstructural latent factor dwiF3					
WM Lesion	0.432	0.113 – 0.752	0.168	2.652	0.010
SDC 40%	0.060	0.005 – 0.115	0.028	2.124	0.043
SDC 60%	0.077	0.022 – 0.133	0.028	2.747	0.011
SDC 80%	0.078	0.006 – 0.150	0.037	2.128	0.043

Microstructural and connectivity association with behavior





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Mapping EEG waves in Stroke patients

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Patient info & Methods

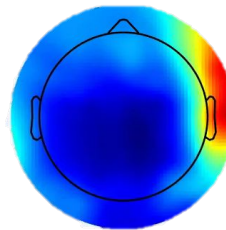
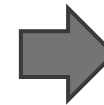
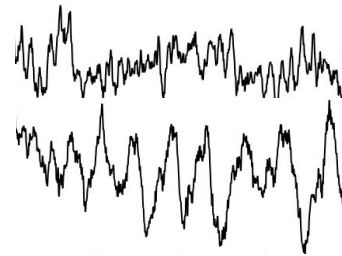
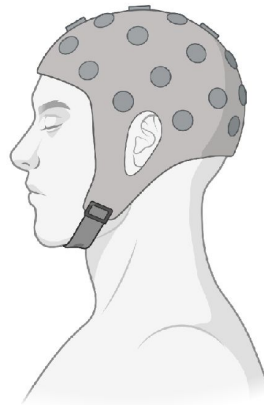
Dataset



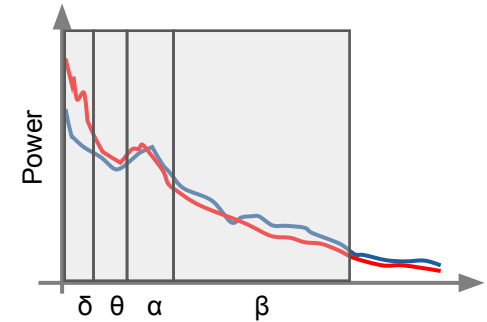
54 stroke patients
(63 ± 14 years;
20 females)

Location	#
20 healthy controls	
Cortical	10
Subcortical	18
Cortical and subcortical	19
Subtentorial	7

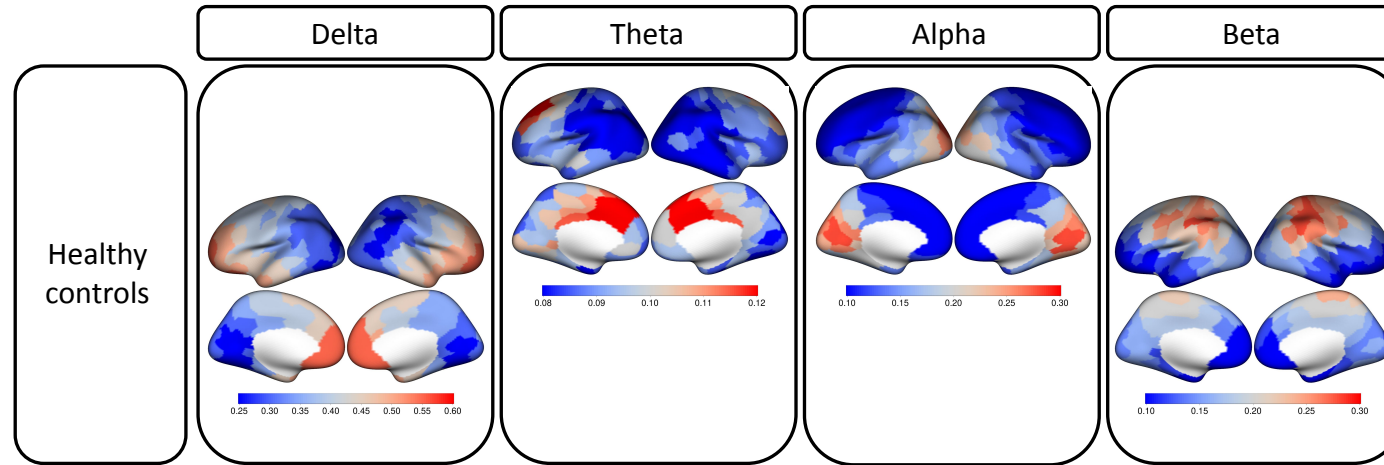
From eeg to source



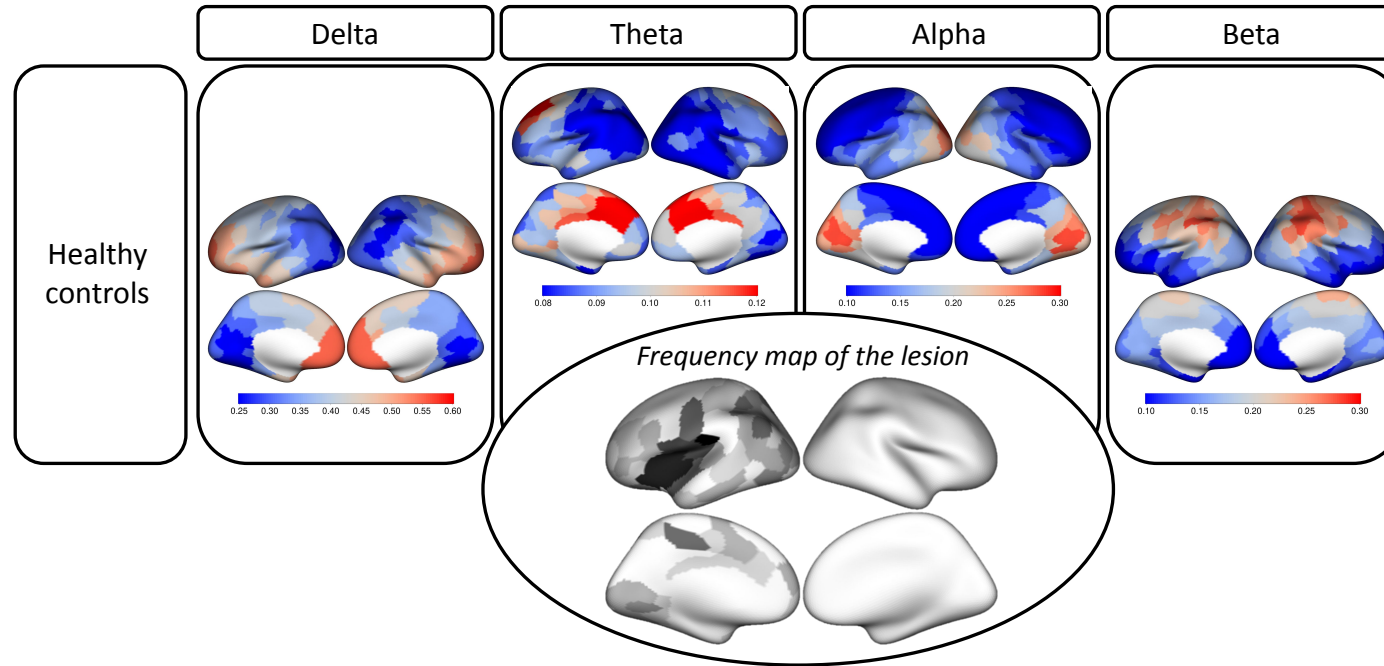
10 min EC
10 min EO



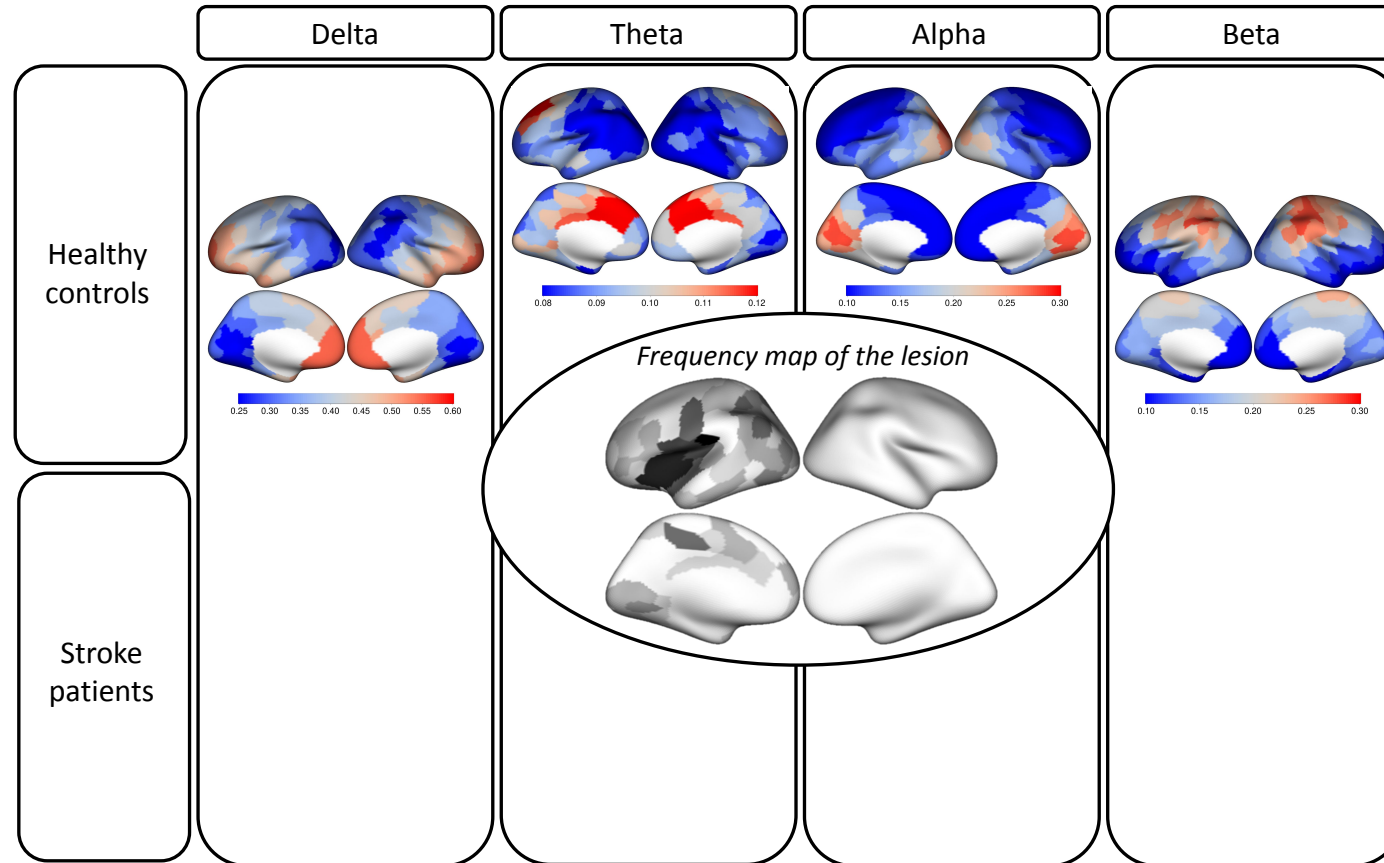
How does the distribution of power change after a stroke?



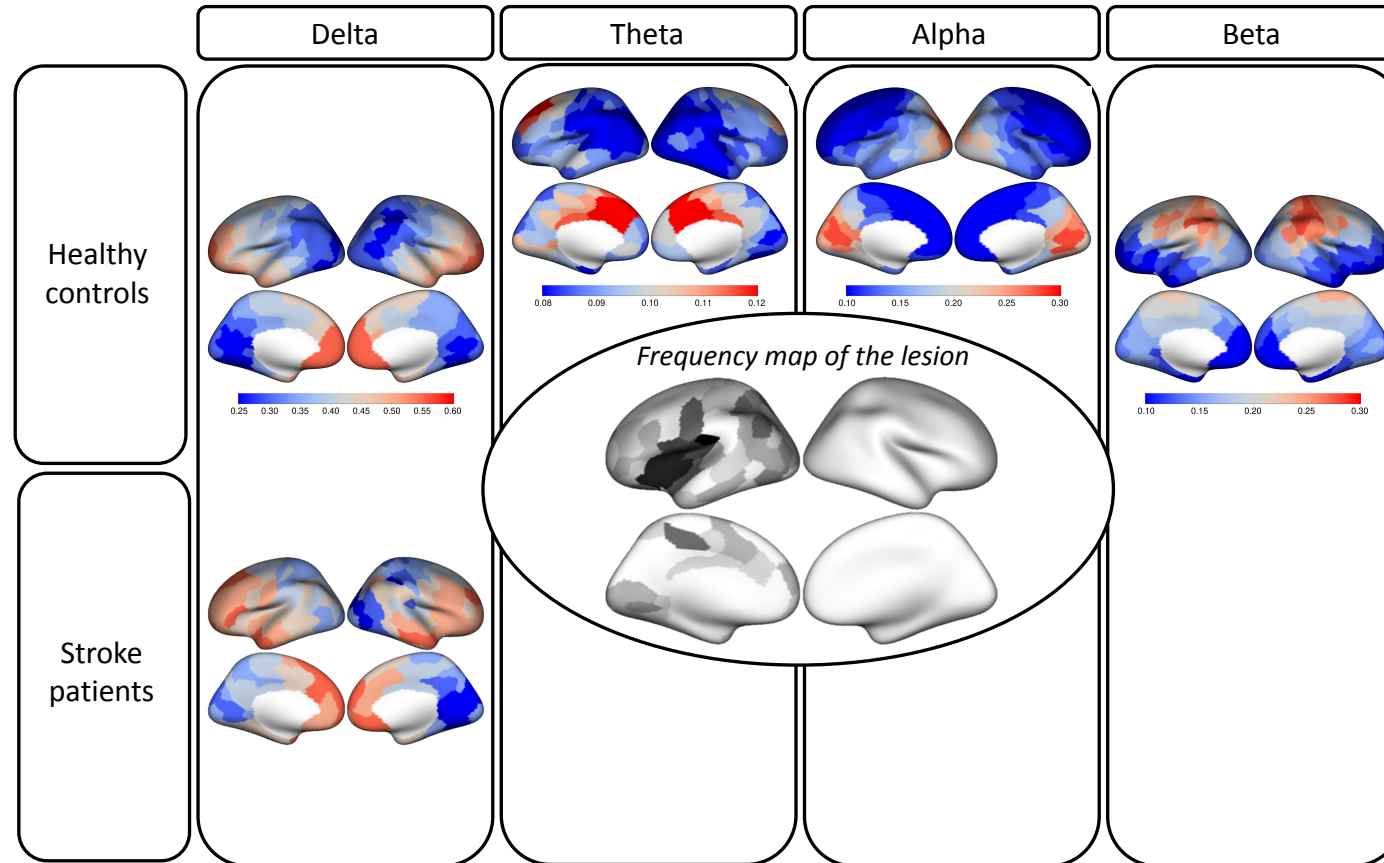
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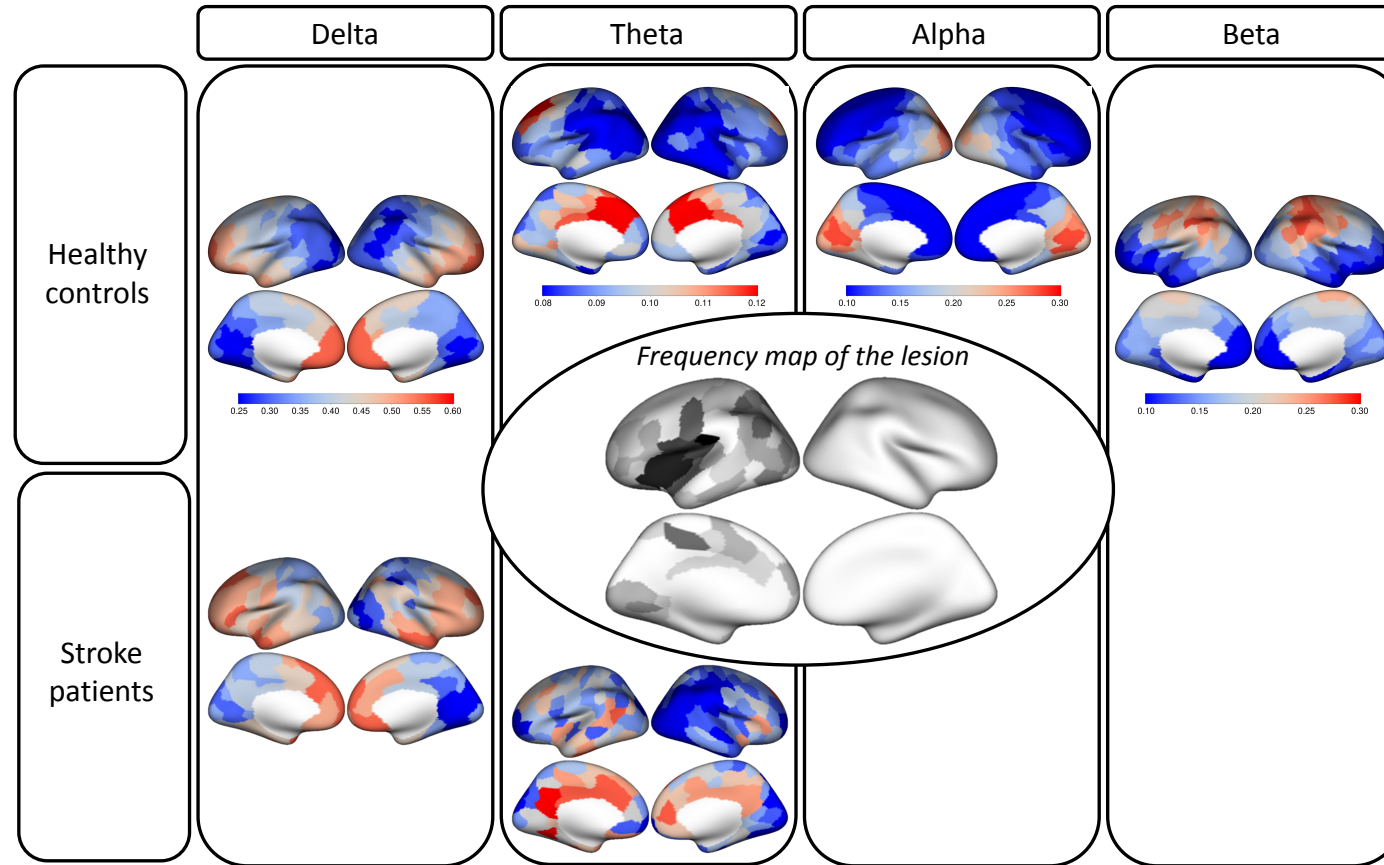
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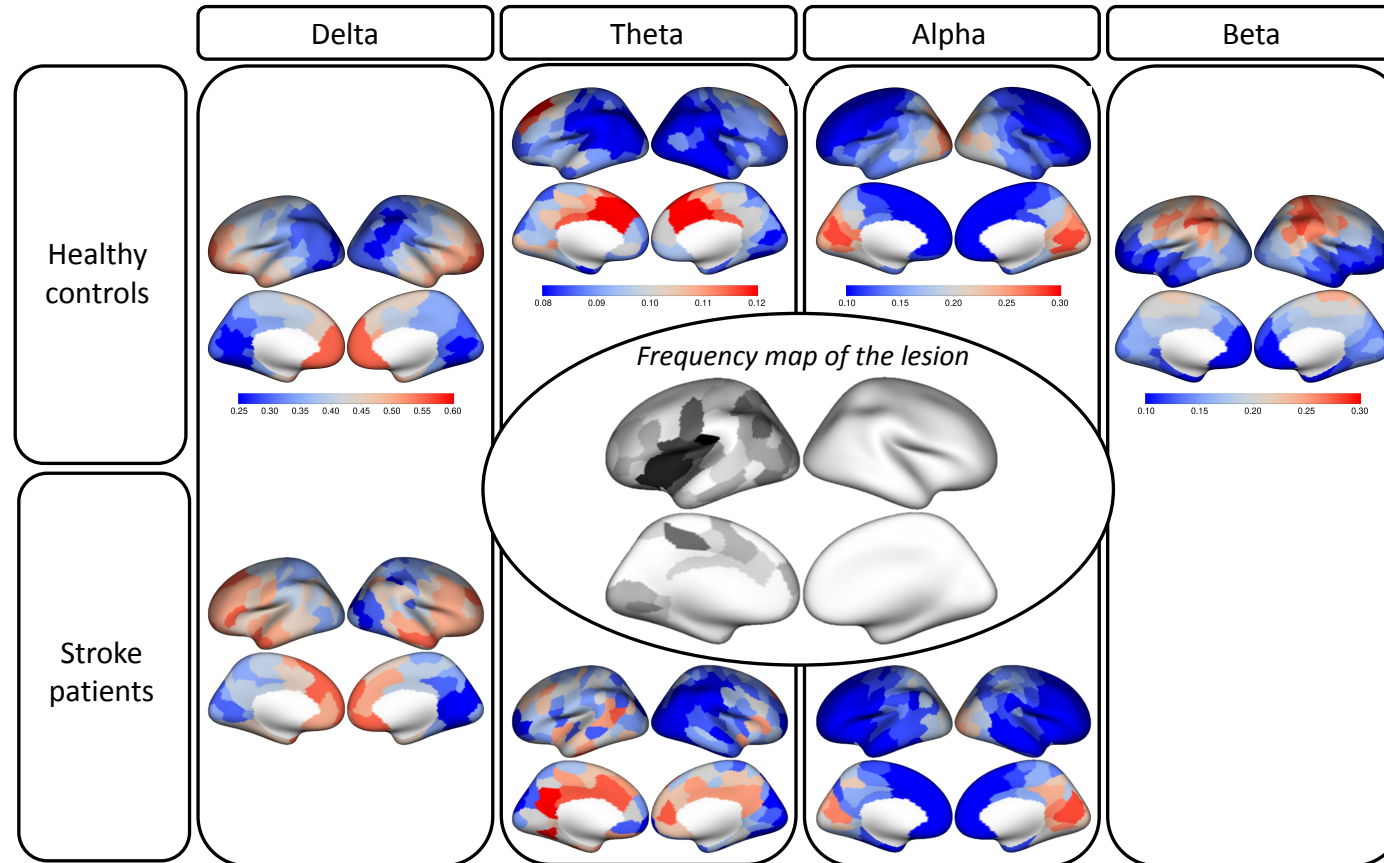
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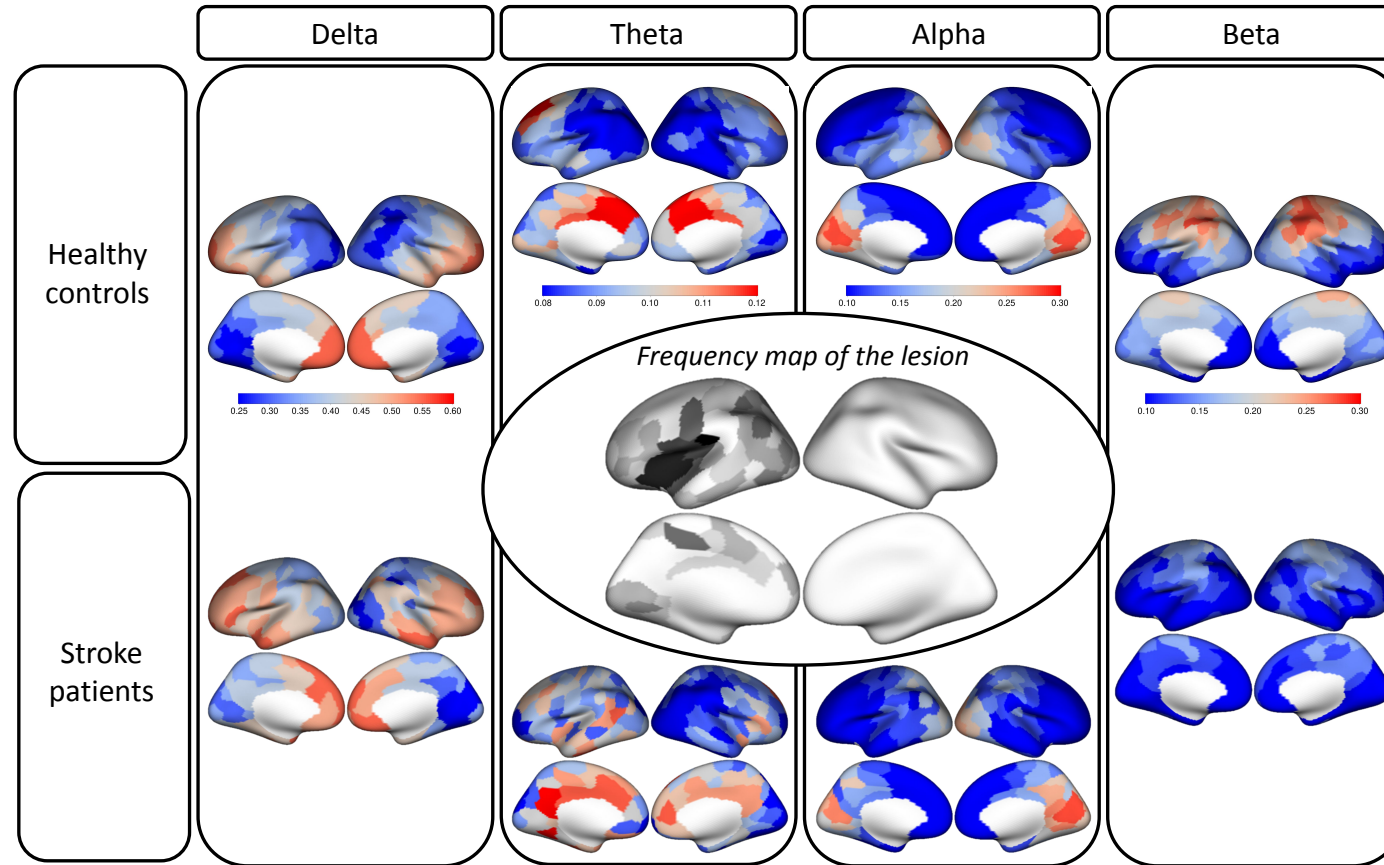
How does the distribution of power change after a stroke?



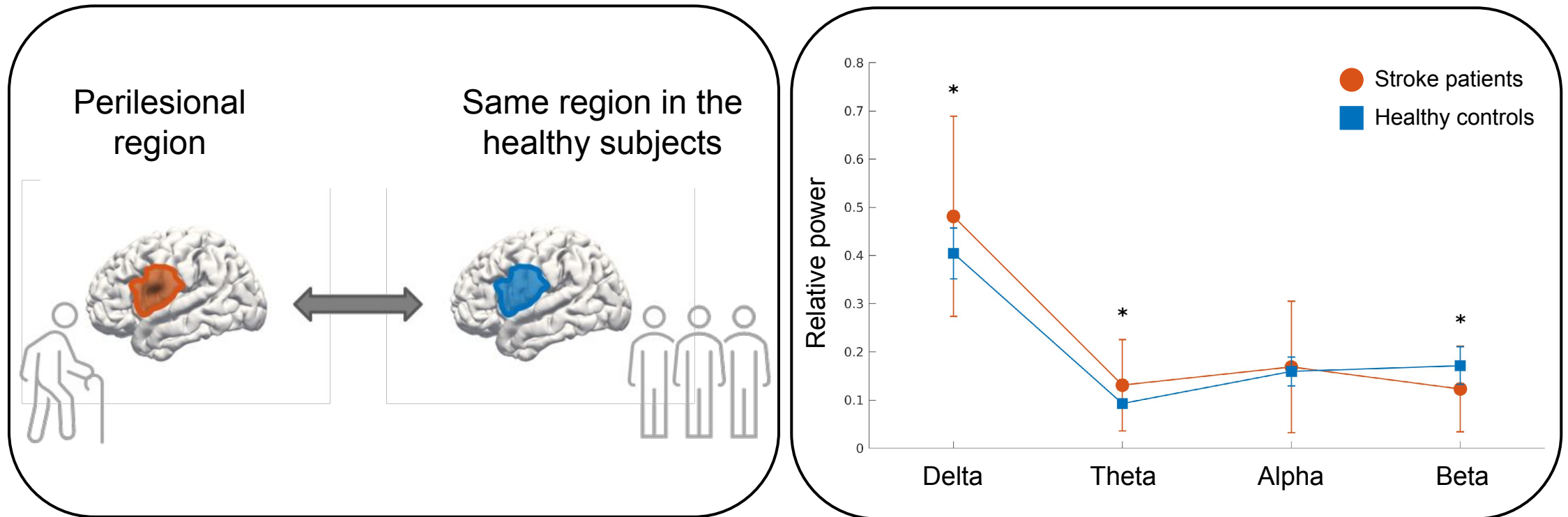
How does the distribution of power change after a stroke?



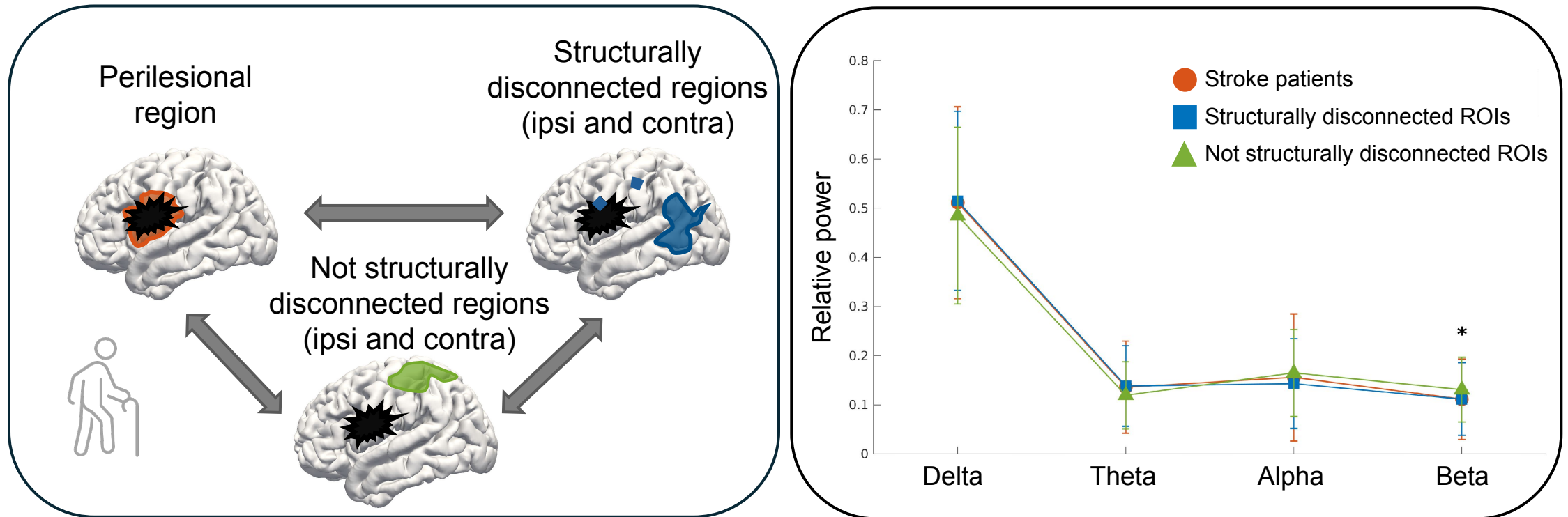
How does the distribution of power change after a stroke?



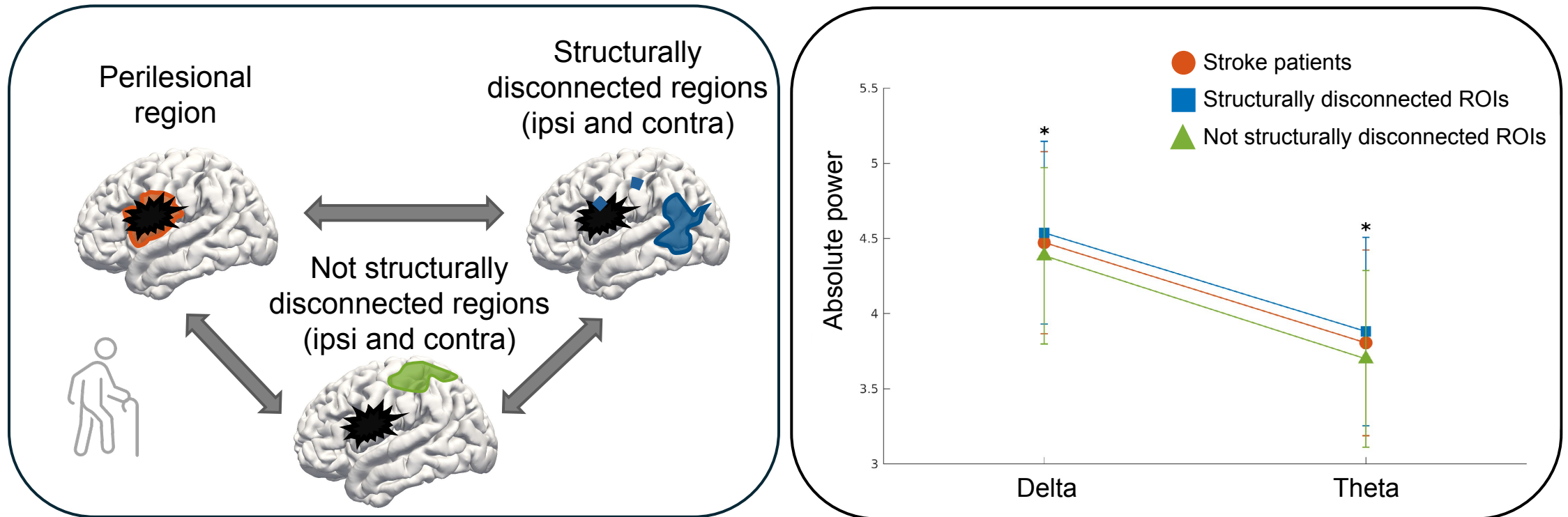
How does the distribution of power change after a stroke?



How does the distribution of power change after a stroke?

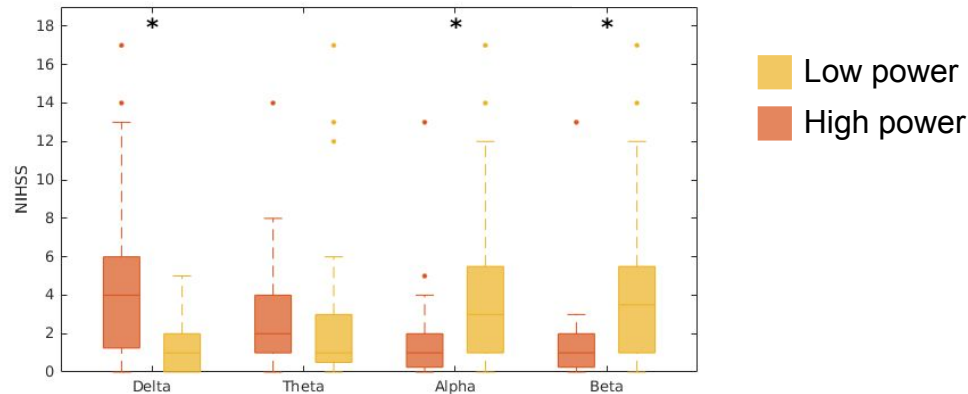
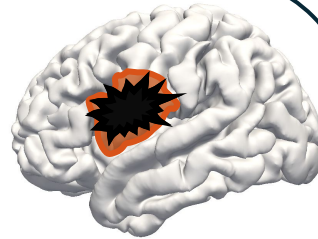


How does the distribution of power change after a stroke?



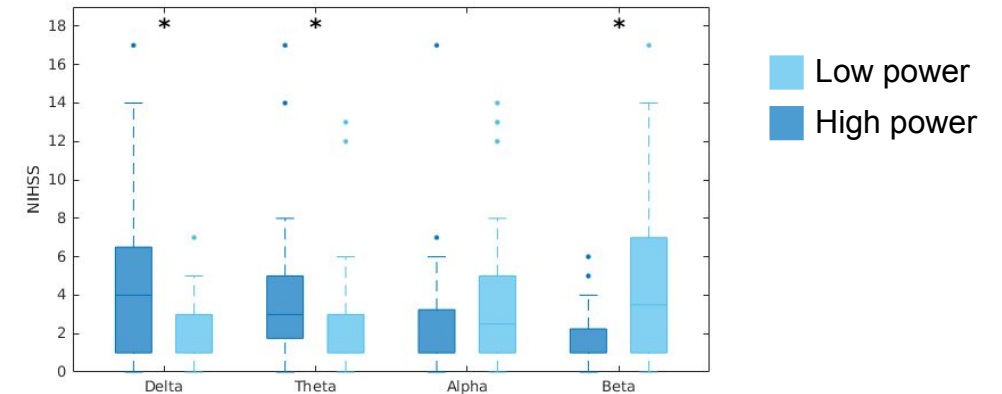
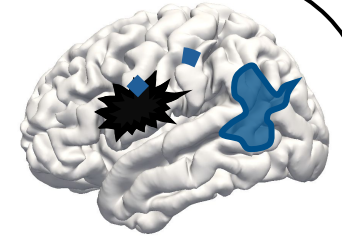
Is the power related to the severity of the stroke?

Relationship with power in the perilesional space



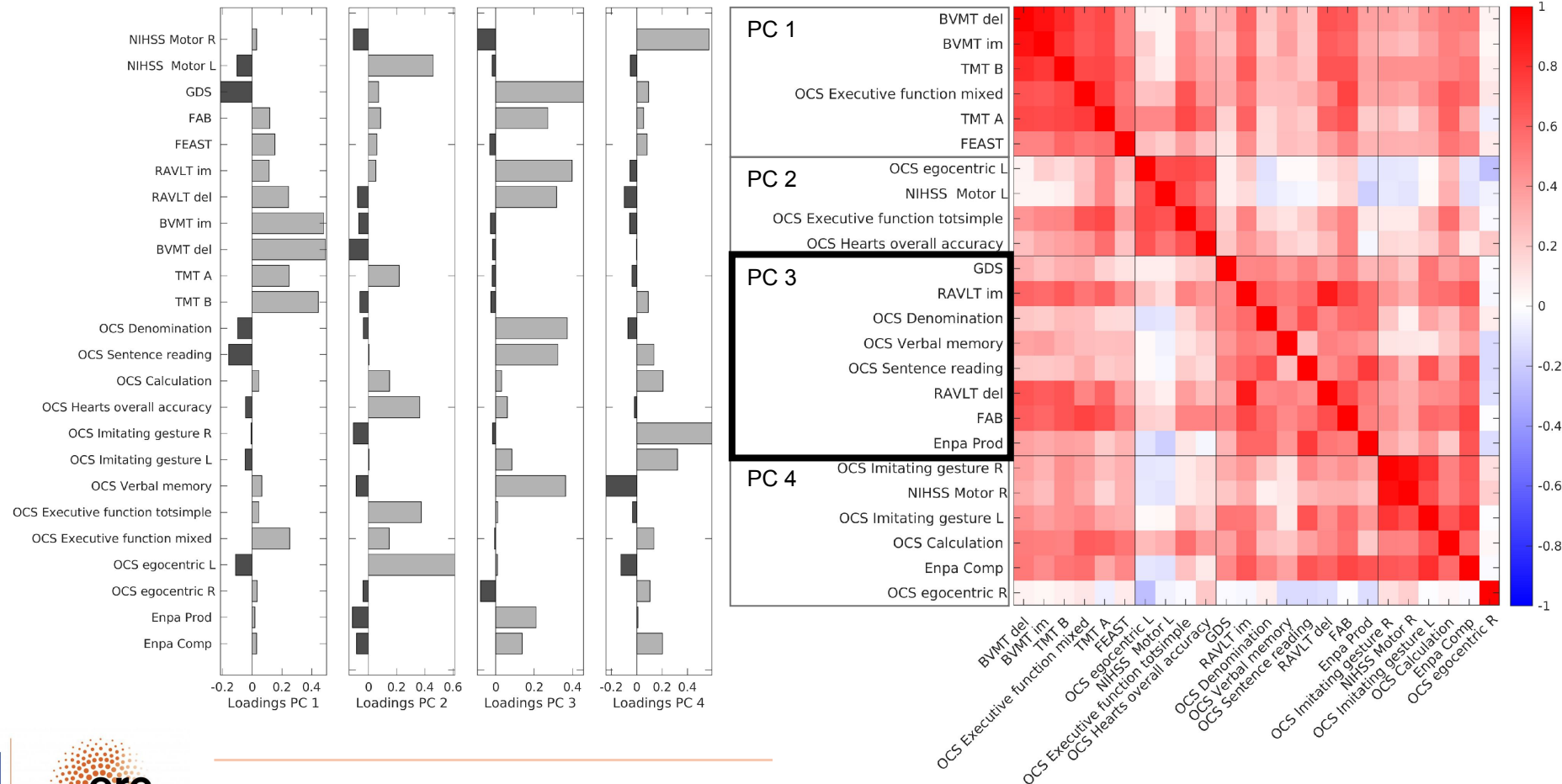
Worse performance:
 Higher Delta
 Lower Alpha
 Lower Beta

Relationship with power in structurally disconnected regions



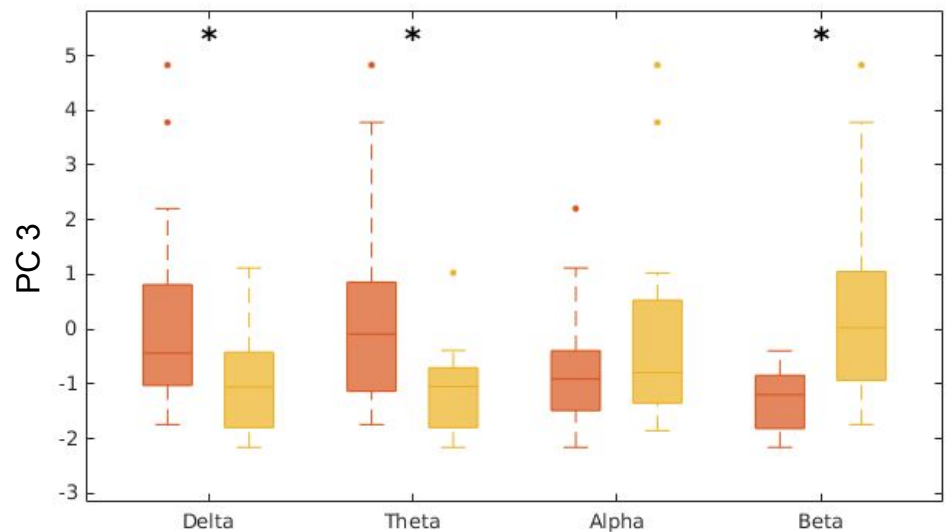
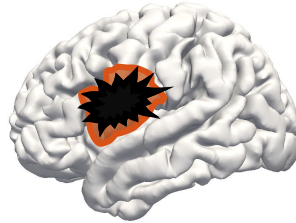
Worse performance:
 Higher Delta
 Higher Theta
 Lower Beta

Is the power related to the behavioural/cognitive profile of the patient?



Is the power related to the behavioural/cognitive profile of the patient?

Relationship with power in the perilesional space



Low power
High power

Worse performance: Higher Delta
 Higher Theta
 Lower Beta

Take home messages

- In the perilesional space, delta power is higher and beta power is lower compared to the same region in healthy subjects.
- Regions that are not structurally disconnected show lower delta power and higher beta power compared to both perilesional and structurally disconnected regions.
- Higher delta power and lower beta power appear to be associated with poorer performance and greater stroke severity.

